

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)

Question Count: 5

Total Marks: 25

Code of Conduct

I _A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _A.Reddy Prasad

1. Cloud Auto-Scaling and Load Balancing

During seasonal sales, an e-commerce company experiences sudden increases in website traffic. **Cloud auto-scaling** automatically adds or removes servers based on demand, while **load balancing** distributes incoming requests evenly across available servers.

Auto-Scaling

- Automatically increases server instances during high traffic.
- Removes extra instances when traffic decreases.
- Prevents server overload.

Load Balancing

- Distributes user requests across multiple servers.
- Improves response time.
- Prevents a single server from becoming overloaded.

Benefits

- Reduces downtime.
- Improves website performance.
- Provides a better user experience.
- Optimizes cloud resource usage and cost.

Role of Monitoring Tools

Monitoring tools continuously track CPU usage, memory, network traffic, and server health. If resource usage exceeds a threshold, they trigger auto-scaling and generate alerts for administrators.

Real-World Example

During the **Amazon Great Indian Festival** or **Flipkart Big Billion Days**, millions of customers visit the website simultaneously. Auto-scaling launches additional servers, while load balancing distributes customer requests efficiently, ensuring smooth shopping without delays.

2. Cloud Storage for Backup Data

Cloud storage provides a secure and reliable solution for storing large volumes of backup data.

Durability and Redundancy

- Data is automatically stored in multiple locations.

- Multiple copies protect against hardware failures.
- Replication ensures data is available even if one server fails.

Security

- **Encryption** protects data during storage and transmission.
- **Access control** allows only authorized users to access backup files.
- Multi-factor authentication (MFA) provides additional security.

Cost Advantages

- No need to purchase expensive storage hardware.
- Pay only for the storage used.
- Reduced maintenance and infrastructure costs.

Disaster Recovery Example

If a company's primary data center is damaged by a flood or fire, backup data stored in the cloud can be restored quickly, allowing business operations to continue with minimal downtime.

3. Content Delivery Network (CDN)

A **Content Delivery Network (CDN)** is a network of servers located around the world that delivers content from the server closest to the user.

Performance Improvement

- Reduces data travel distance.
- Loads websites and applications faster.
- Decreases network latency.

Edge Servers

- Store cached copies of frequently accessed content.
- Deliver data from nearby locations.
- Reduce the load on the main server.

Global Availability

Cloud providers deploy servers in multiple regions worldwide. Users are automatically connected to the nearest available server, ensuring fast and reliable service.

Example

Streaming platforms like **Netflix** or online games use CDNs to deliver videos and game data quickly to users worldwide, providing smooth streaming and low-latency gameplay.

4. Cloud Security and Compliance

Financial institutions require strong security and compliance when using cloud services.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI-DSS
- GDPR
- SOC 2

These standards help organizations meet legal and security requirements.

Logging and Monitoring

- Record all user activities.
- Detect suspicious behavior.
- Help during security audits and investigations.

Encryption

- Encrypts data at rest and in transit.
- Protects sensitive financial information from unauthorized access.

Data Sovereignty

Organizations can choose cloud regions where data is stored to comply with local government regulations and privacy laws.

Example

During an online banking transaction, customer information and payment details are encrypted, all activities are logged, and the data is stored in an approved regional data center to meet compliance requirements.

5. Containers and Kubernetes

Containers package an application and its dependencies so it can run consistently across different environments.

Containers vs Virtual Machines

Containers

Lightweight

Faster startup

Share host OS

Lower resource usage

Virtual Machines

Heavier

Slower startup

Separate operating system

Higher resource usage

Kubernetes Orchestration

Kubernetes simplifies:

- Automatic deployment
- Load balancing
- Auto-scaling
- Self-healing of failed containers
- Centralized management

Portability

Containers run consistently across different cloud platforms such as AWS, Microsoft Azure, and Google Cloud, reducing compatibility issues.

Example

An **online shopping application** can use microservices such as:

- User Service
- Product Service
- Order Service
- Payment Service
- Notification Service